

SciPIP: An LLM-based Scientific Paper Idea Proposer

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Abstract

The rapid advancement of large language models (LLMs) has opened new possibilities for automating the proposal of innovative scientific ideas. This process involves two key phases: literature retrieval and idea generation. However, existing approaches often fall short due to their reliance on keyword-based search tools during the retrieval phase, which neglects crucial semantic information and frequently results in incomplete retrieval outcomes. Similarly, in the idea generation phase, current methodologies tend to depend solely on the internal knowledge of LLMs or metadata from retrieved papers, thereby overlooking significant valuable insights contained within the full texts. To address these limitations, we introduce SciPIP, an innovative framework designed to enhance the LLM-based proposal of scientific ideas through improvements in both literature retrieval and idea generation. Our approach begins with the construction of a comprehensive literature database that supports advanced retrieval based not only on keywords but also on semantics and citation relationships. This is complemented by the introduction of a multi-granularity retrieval algorithm aimed at ensuring more thorough and exhaustive retrieval results. For the idea generation phase, we propose a dual-path framework that effectively integrates both the content of retrieved papers and the extensive internal knowledge of LLMs. This integration significantly boosts the novelty, feasibility, and practical value of proposed ideas. Our experiments, conducted across various domains such as natural language processing and computer vision, demonstrate SciPIP’s capability to generate a multitude of innovative and useful ideas. These findings underscore SciPIP’s potential as a valuable tool for researchers seeking to advance their fields with groundbreaking concepts.

1. Introduction

With the rapid expansion of scientific knowledge and the increasing complexity of interdisciplinary research, researchers face significant challenges, including information extraction and knowledge synthesis when generating novel ideas. The advent of large language models (LLMs), such as GPT-4 (Ouyang et al., 2022), LLaMA (Touvron et al., 2023a;b), Qwen (Bai et al., 2023; Yang et al., 2024), GLM-4 (Zeng et al., 2024), DeepSeek V3 (DeepSeek-AI et al., 2024), and others, provides promising solutions. These models can process vast amounts of literature at speeds far beyond human capabilities, effectively mitigating concerns about information overload. Moreover, their broad interdisciplinary knowledge enables them to synthesize insights across domains, fostering the generation of innovative ideas.

Recent research has explored the potential of LLMs in assisting researchers with idea generation. Most approaches involve two phases: *literature retrieval* and *idea generation*. Despite their promise, both phases have limitations: (1) *During the literature retrieval phase*, most existing algorithms (Lu et al., 2024; Radensky et al., 2024) rely on keyword-based retrieval techniques, which fail to leverage semantic information inherent in textual data, significantly compromising retrieval thoroughness. While some approaches employ semantic information (Su et al., 2024; Xu et al., 2023) through vector-based retrieval, they often encode entire sections (e.g., abstracts) into vectors. However, as each section typically contains multifaceted information, such an approach makes it difficult to capture key points effectively. This, in turn, impacts both encoding quality and retrieval performance. (2) *During the idea generation phase*, existing idea generation approaches (Lu et al., 2024; Baek et al., 2024; Wang et al., 2024a) mainly rely on the LLM’s internal knowledge and do not take full advantage of retrieved papers.

To address these challenges, we propose SciPIP, a novel framework designed to enhance idea proposal. In particular, for a more thorough literature retrieval, we first **construct a literature database**. Specifically, we collect approximately 78K papers from several top-tier academic conferences of AI field and utilize LLMs to re-summarize each paper into a structured quintuple consisting of keywords, backgrounds,

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ideas, concise methods, and references. These components are individually preprocessed (e.g., encoded into vectors) and stored in a database to enable more precise and efficient retrieval. Based on our constructed database, we propose a **multi-granularity retrieval method**, which comprehensively leverages keywords, semantic embeddings and citation relations, enabling a thorough literature retrieval across multiple dimensions.

In the idea generation phase, we propose a **dual-path idea generation framework**. This approach explicitly separates idea generation into two pathways: one leveraging the LLM’s internal knowledge and the other utilizing retrieved literature. The LLM is then guided to integrate these two pathways, synthesizing valuable insights from both to generate comprehensive and well-informed final ideas.

Extensive experiments are conducted to evaluate both idea proposal and literature retrieval on the Natural Language Processing (NLP) and Computer Vision (CV) domains. Various idea generation tools including SciPIP propose ideas based on a given user’s query, and the quality of the proposed ideas are assessed by human experts. Results show that SciPIP significantly outperforms existing idea generation approaches in both CV and NLP domains in terms of novelty, feasibility, clarity, relevance, *etc.*

The main contributions of this paper include:

- We construct a literature database for AI papers, with their key information and embeddings extracted, which can benefit the community for future research.
- Based on the literature database, we introduce a multi-granularity literature retrieval algorithm.
- We present an LLM-based dual-path idea generation framework, improving the proposed research ideas in terms of the novelty, clarity, feasibility, *etc.*
- Extensive experiments are conducted to demonstrate the effectiveness of SciPIP.

2. Related Works

LLM-based Scientific Idea Proposal. Some previous works (Radensky et al., 2024; Su et al., 2024; Xu et al., 2023; Hu et al., 2024; Lu et al., 2024; Baek et al., 2024; Wang et al., 2024a) have explored how to generate research ideas using large language models. These methods typically involve literature retrieval and idea generation. For instance, AI Scientist (Lu et al., 2024) relies on keyword-based search engines such as Semantic Scholar for literature retrieval, which fail to leverage the semantic information of the documents. In contrast, (Su et al., 2024) employs a FAISS (Johnson et al., 2019) vector search database, where each vector is encoded from full titles, abstracts, and other metadata. Some approaches such as (Baek et al., 2024) also

utilize the citation lists of papers to select relevant references. In the idea generation phase, (Lu et al., 2024) relies on the internal knowledge of the model to propose ideas. Additionally, AI Scientist refines repetitive ideas through prompt engineering after performing a post-generation literature search, aiming to enhance the novelty of the final ideas. ResearchAgent (Baek et al., 2024), on the other hand, focuses on identifying problems in the retrieved papers and suggesting corresponding solutions. However, these algorithms primarily utilize only the abstracts of the retrieved papers, often resulting in a lack of necessary details in their methods. In SciPIP, we re-summarize each paper into a structured quintuple.

Retrieval Augmented Generation. Many retrieval-augmented generation (RAG) algorithms have been proposed to enhance performance on various retrieval-required downstream tasks. Typical RAG methods involve three primary steps: indexing, retrieval, and generation (Gao et al., 2023b). Several mature RAG algorithms have demonstrated effectiveness across diverse applications (Wang et al., 2024b; Peng et al., 2024; Gao et al., 2023a; Yu et al., 2023; Shao et al., 2023; Wang et al., 2023). These methods rely on a combination of retrieval techniques such as DPR and LLMs to generate high-quality, context-sensitive results. Scientific idea proposal can be seen as a special case of retrieval-augmented generation. However, it differs from existing RAG approaches in some key aspects. For example, most RAG algorithms perform retrieval based on relevance, where the query is matched to the keys in the database. However, for idea generation, the number of documents relevant to a specific domain background is often excessively large, making it prone to introducing redundant literature.

Text Embedding Models. Text embedding models Reimers & Gurevych, 2019; Wang et al., 2022 are designed to represent textual data as high-dimensional vectors, capturing their semantic meaning in a machine-readable format. These embeddings are widely used in various tasks, including text classification, information retrieval, and text matching. Different tasks often require tailored embedding strategies to achieve optimal performance. For instance, embeddings for information retrieval (Lee et al., 2024; Sturua et al., 2024; Chen et al., 2024; Zhang et al., 2024; Neelakantan et al., 2022) are typically asymmetric, meaning that the query and the text being retrieved are encoded using different models or prompts. In this case, a high similarity between the query and the retrieved passage indicates strong relevance. On the other hand, embeddings (Reimers & Gurevych, 2019; Sturua et al., 2024; Neelakantan et al., 2022) for text matching are symmetric, where both text segments are encoded in the same manner. The similarity score reflects the semantic equivalence between the two text segments.

In addition, we also investigate off-the-shelf literature re-

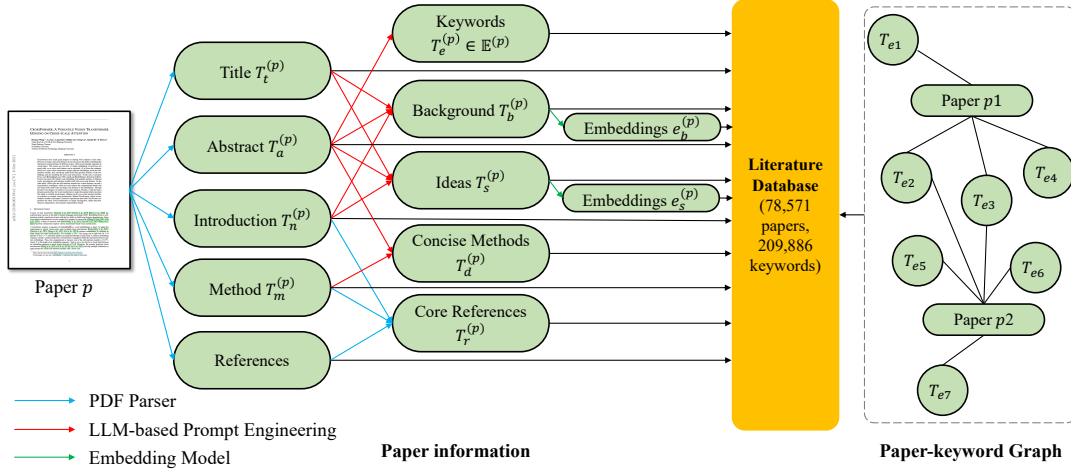


Figure 1. The pipeline of constructing the literature database. Paper sections are extracted via a PDF parser, summarized by an LLM, encoded, and stored in the database. A paper-keyword graph linking each paper to its keywords is also stored.

retrieval tools. These details are provided in Appendix A due to space limitations.

3. Methods

We introduce a scientific idea proposer called SciPIP, which comprises two key phases: literature retrieval and idea generation. Since existing retrieval algorithms and literature databases are not directly applicable to SciPIP, we first construct a database in Section 3.1. This process involves summarizing the key information from each paper and encoding these summaries into embeddings. Subsequently, we elaborate on our multi-granularity literature retrieval approach in Section 3.2. Finally, inspired by the way humans approach research, we present a dual-path idea generation framework in Section 3.3, which leverages the retrieved literature and the internal knowledge of large language models (LLMs) to propose innovative and feasible ideas.

3.1. Literature Database Construction

General literature retrieval either relies on keyword matching or encodes sections of papers or fixed-length chunks to build an index. However, the former overlooks semantic information, and the latter introduces significant noise. We argue that for scientific retrieval aimed at supporting idea proposal, the focus should be on the background, including the problems or challenges within a research field, as well as the ideas or methods presented in the target literature. Therefore, by summarizing the literature in advance and extracting the most relevant information for idea proposal to build the index, the effectiveness of literature retrieval can be significantly improved.

To this end, as illustrated in Figure 1, we compile a dataset

using papers published in the past decade from several top-tier academic conferences. A PDF parser is then employed to extract titles, abstracts, introductions, methods, and references. Considering that the background of a paper is typically described in the title, abstract, and introduction, we use these sections as input and design a prompt template. By leveraging prompt engineering, we extract the background and ideas information effectively. Similarly, another two prompt templates are developed to summarize the keywords and concise methods of the paper. Formally,

$$\begin{aligned} (T_b^{(p)}, T_s^{(p)}) &= f(\tau_1, T_t^{(p)}, T_a^{(p)}, T_n^{(p)}), \\ \mathbb{E}^{(p)} &= f(\tau_2, T_a^{(p)}), \\ T_d^{(p)} &= f(\tau_3, T_m^{(p)}), \end{aligned} \quad (1)$$

where f , $T_t^{(p)}$, $T_a^{(p)}$, $T_n^{(p)}$, $T_m^{(p)}$ are the LLM, the paper p 's title, abstract, introduction, and method sections, respectively. $\mathbb{E}^{(p)}$, $T_b^{(p)}$, $T_s^{(p)}$, $T_d^{(p)}$, $T_r^{(p)}$ are extracted keywords, background, ideas, concise methods, and core references, as shown in Figure 1. $\tau_i, i \in \{1, 2, 3\}$ represent our designed prompt templates, and details of these prompt templates are provided in Table 6. After summarization, we encode the extracted background and ideas into embeddings using a text-matching embedding model and construct the retrieval index, as shown in Equation (2).

$$e_b^{(p)} = g(T_b^{(p)}), e_s^{(p)} = g(T_s^{(p)}), \quad (2)$$

where $e_*^{(p)}$ represents embeddings, which are generated offline and stored in the database, ensuring efficient retrieval.

Besides, to enhance retrieval efficiency, we construct a paper-keyword graph within the database, capturing the relationships between papers and keywords. Specifically,